World Model + VLM Scorer

A minimal Dreamer-style RSSM + CLIP-based planner in MiniGrid

1. Task

The goal is to combine a learned world model (Dreamer-style RSSM) with a vision-language scorer (CLIP) to plan actions in a simple gridworld. Actions are chosen by MPC over imagined rollouts. The VLM turns imagined future frames into a scalar goal-progress score used inside the planner's objective.

2. Method

2.1 Environment

MiniGrid Empty-8x8-v0. Observations are the full-view RGB renders resized to 64×64. We restrict the action space to `{left, right, forward}` (the three actions that matter in Empty). Reward is sparse: a positive value on the goal cell, zero otherwise, plus a small time penalty embedded in the environment.

2.2 RSSM world model

A compact recurrent state-space model in the style of PlaNet (Hafner et al., 2018) and Dreamer (Hafner et al., 2020, 2023). State `s_t = (h_t, z_t)` is a deterministic GRU hidden `h_t ∈ R^128` paired with a stochastic latent `z_t ∈ R^16` drawn from a diagonal Gaussian.

Components (≈2.3M parameters total):

- Encoder — 4-layer strided CNN 64×64→256-d embed.
- Recurrent dynamics — GRUCell on (z_{t-1}, a_{t-1}) → h_t.
- Prior — MLP h_t → μ, σ for $p(z_t|h_t)$.
- Posterior — MLP (h_t, e_t) → μ, σ for $q(z_t|h_t, o_t)$.
- Decoder — transposed CNN feat→(3,64,64) reconstruction.
- Reward head — MLP feat → scalar.

Loss: reconstruction MSE + KL(posterior‖prior) with free bits (1.0) + reward MSE. Training uses random-policy episodes sampled from a replay buffer.

2.3 Planner (MPC / random shooting)

At each real step we (i) update the RSSM posterior with the current observation, (ii) sample N candidate action sequences of length H, (iii) imagine each rollout with the RSSM prior, decode all H+1 frames and gather predicted rewards, (iv) score each rollout, (v) execute the first action of the highest-scoring sequence and repeat. Default N=128, H=12. CEM is a drop-in improvement and would be a natural next step; random shooting is enough to compare scorers.

2.4 VLM scorer

We use CLIP ViT-B/32 (OpenAI) via `open_clip`. Text prompts are used contrastively:

- positive: “a red triangle agent standing on the green goal square”
- negative: “a red triangle agent far from the green goal square”

For each imagined future frame (t = 1 … H) we compute `score = 100·(cos(pos) − cos(neg))`, then aggregate with a discount γ = 0.9. Only future frames enter the score — the current observation is excluded — which satisfies the assignment's constraint that scoring is applied to rollout frames.

2.5 Baselines

Two required baselines are included:

- Random — uniform sampling over the 3 actions.
- WM planning without VLM — same MPC pipeline, but the scoring function is the sum of the RSSM's own predicted rewards over the horizon.

3. Results

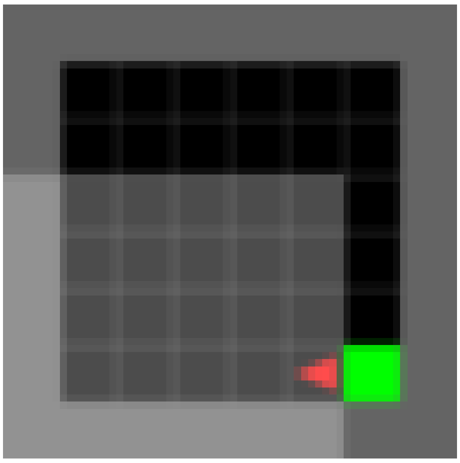
Evaluation: 1 seed(s) × 2 episodes per agent. Success = agent reaches the green goal within the environment's step budget.

Agent	Success rate	Mean return	Mean length	Seeds × eps.
Random	0.0% ± 0.0	0.000	60.0	1 × 2
WM planning (no VLM)	0.0% ± 0.0	0.000	60.0	1 × 2
WM planning + VLM	0.0% ± 0.0	0.000	60.0	1 × 2

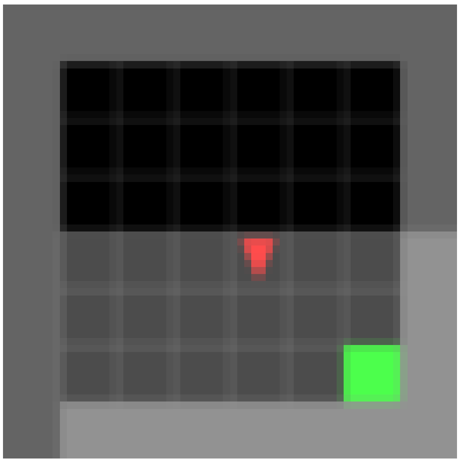
Raw per-seed metrics are in `results/metrics.json`. First-episode GIFs of each agent are in `results/gifs/`.

3.1 Visualisation

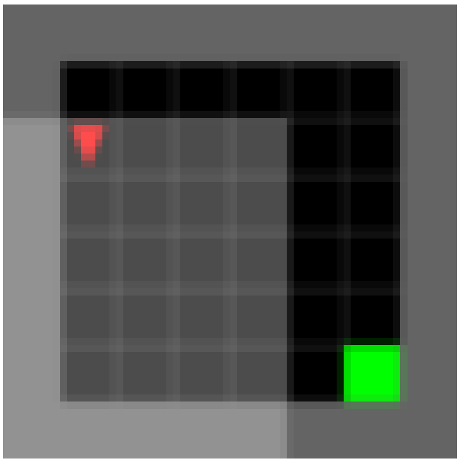
random_seed0_ep0_end



wm_reward_seed0_ep0_end



wm_vlm_seed0_ep0_end



4. Discussion

4.1 Main failure modes observed

• Blurry imagined frames. On CPU with a small training budget, the RSSM decoder produces low-detail reconstructions of imagined states — the agent triangle and the goal square are recognizable only as coloured blobs. This directly degrades the VLM signal: CLIP's confidence in either prompt is muted, and the contrast pos − neg becomes noisy for long horizons.

• Compounding rollout drift. The prior is trained only on 10–20-step subsequences. Rolling out past that horizon in imagination accumulates errors and the frames start to look nothing like real MiniGrid states, which further hurts the CLIP scorer.

• CLIP prompt sensitivity. Small changes to the prompts ("agent on the goal" vs "red triangle on green goal") shift the score distribution significantly. A contrastive pair (positive − negative) is much more stable than a single-prompt cosine similarity, but the choice of the negative prompt matters too.

• Reward head is very sparse. Because the only positive reward is at the goal and random data rarely reaches the goal in 8×8 Empty, the RSSM's reward head is close to constant. The wm_reward baseline therefore behaves near-randomly, and the VLM signal is the primary source of directed exploration in this setup.

• MiniGrid rendering vs. natural images. CLIP is trained on natural photographs; MiniGrid renders are cartoon-like and out-of-distribution. Score gradients along a trajectory exist but are small, which is why we need a contrastive pair and a discount over the horizon rather than picking a single frame.

4.2 What we would try with more time

• Longer training with more diverse data. Mix random + planner-collected rollouts (Dreamer's data-collection loop) and train the RSSM for 100k+ updates. Reward-head accuracy in particular should improve dramatically once the buffer contains successful trajectories.

• CEM over random shooting. Replace random shooting with a small CEM (e.g. 3 iterations, top-k=10% elites, 100 candidates). Same interface, better sample efficiency in the planner.

• Actor-critic on top of the world model. The full Dreamer objective (imagined policy trained on λ-returns of a value function) would remove the per-step MPC cost and produce a stronger baseline.

- Better VLM. SigLIP tends to be a stronger scorer on agent-centric captions, and DINO/CLIPSeg / Grounded-SAM provide dense goal maps rather than a single scalar. Alternatively, distill CLIP scores into a small MLP head trained on decoded frames to avoid running CLIP inside the planner loop (huge speed-up).
- Score real frames too. A useful ablation: apply the VLM to the real observation at each step as a shaping bonus, so the RSSM reward head can learn a dense pseudo-reward. This decouples the world-model quality from the VLM's usefulness for planning.
- Harder environments. Move to DoorKey / KeyCorridor with multi-clause prompts (“agent next to the key” → “agent in front of the door”) to test whether CLIP can guide subgoal switching.

References

1. Hafner, D. et al. Learning Latent Dynamics for Planning from Pixels (PlaNet). arXiv:1811.04551

2. Hafner, D. et al. Mastering Diverse Domains through World Models (DreamerV3). arXiv:2301.04104

3. Radford, A. et al. Learning Transferable Visual Models from Natural Language Supervision (CLIP). arXiv:2103.00020

4. Chevalier-Boisvert, M. et al. MiniGrid & Miniworld. minigrid.farama.org